Uniformly stable frozen potentials can generate an exponentially unstable parabolic system

Candidate partial result for arXiv:2208.10324

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Status. Candidate partial result, likely valid, pending human review. The result is a complete negative theorem for pointwise spectral stability as a boundedness criterion. It does not characterize all non-dissipative potentials.

1 Source problem and scope

Dobrick and Glück study Neumann parabolic systems coupled by a matrix-valued potential [1]. In their concluding remarks (source PDF, page 16) they observe that, outside potentials dissipative for an ℓ^p-norm, it is not clear in general how to check boundedness of the solutions. The source passage is reproduced below.

> *Concluding remarks.* Clearly, much more remains to be done in the case that the potential V is not dissipative, since most methods presented in this paper do not work in this case. In fact, it is not even clear to the authors in general how to check boundedness of the solutions to (2.2) if V is not dissipative with respect to any ℓ^p-norm on $\mathbb{C}^N$ (for one exception, though, see Proposition 3.3).
>
> Another direction of generalization is led by the idea to consider compact Riemannian manifolds in place of the bounded domain $\Omega \subseteq \mathbb{R}^d$.
>
> Finally, in view of the coupled first order equations considered in [11], the question arises what happens, in general, if non-elliptic differential operators are coupled by a matrix-valued potential.

The broad request does not prescribe a single candidate criterion. This packet settles the strongest naive one negatively: even if every frozen matrix has the same strictly negative spectrum and its matrix semigroup is uniformly exponentially stable, the spatially coupled PDE can be exponentially unstable.

2 Proof intuition

Choose a smooth Neumann vector field $u(x) = (1, \cos x, \cos 2x)^T$. We want it to be an eigenfunction with any prescribed positive eigenvalue λ. Starting from the stable scalar matrix $-aI$, the missing forcing is $w = (a + \lambda)u - u''$. At every point, u and w are linearly independent. Hence one can choose a smooth linear functional η^T satisfying $\eta^T u = 1$ and $\eta^T w = 0$. The rank-one matrix $N = w\eta^T$ then sends u to w but satisfies $N^2 = 0$. Consequently $V = -aI + N$ retains the frozen spectrum $\{-a\}$, while $u'' + Vu = \lambda u$. Spatial variation in the nilpotent directions is therefore invisible to every pointwise eigenvalue but creates an exact growing PDE mode.

3 The obstruction theorem

Theorem 1. *Let $a > 0$ and $\lambda > 0$. There exists a bounded smooth real potential $V\colon [0,\pi] \to \mathbb{R}^{3\times 3}$ with the following properties.*

1. *For every $x \in [0,\pi]$,*
$$\sigma(V(x)) = \{-a\}$$
with algebraic multiplicity three.

2. *The frozen semigroups are uniformly exponentially stable: for every $\beta < a$ there is $C_\beta < \infty$ such that*
$$\sup_{x\in[0,\pi]} \|\mathrm{e}^{tV(x)}\| \le C_\beta \mathrm{e}^{-\beta t} \qquad (t \ge 0).$$

3. *The Neumann system*
$$\partial_t U = \partial_x^2 U + V(x)U, \qquad \partial_x U(t,0) = \partial_x U(t,\pi) = 0,$$
has the classical solution
$$U(t,x) = \mathrm{e}^{\lambda t}\begin{pmatrix} 1 \\ \cos x \\ \cos 2x \end{pmatrix}.$$
In particular, its solution semigroup is unbounded in every L^p norm, $1 \le p \le \infty$, and in the uniform norm.

Proof. Put
$$u(x) = \begin{pmatrix} 1 \\ \cos x \\ \cos 2x \end{pmatrix}, \qquad w(x) = (a+\lambda)u(x) - u''(x).$$
Write $c = \cos x$ and $d = \cos 2x$. Since
$$u = (1, c, d)^T, \qquad u'' = (0, -c, -4d)^T,$$
we have
$$u \times w = -u \times u'' = (3cd, -4d, c)^T.$$
It follows that
$$D(x) := \|u\|^2\|w\|^2 - (u^T w)^2 = \|u \times w\|^2 = c^2 + (9c^2+16)d^2 > 0. \tag{1}$$
Indeed, c and d cannot vanish simultaneously, since $d = 2c^2 - 1$. Define the smooth column vector
$$\eta(x) = \frac{\|w(x)\|^2 u(x) - (u(x)^T w(x))w(x)}{D(x)}.$$
The Gram-determinant identity (1) gives
$$\eta^T u = 1, \qquad \eta^T w = 0.$$
Set
$$N(x) = w(x)\eta(x)^T, \qquad V(x) = -aI_3 + N(x).$$
The denominator D is positive on the compact interval, so V is smooth and bounded. Moreover,
$$N^2 = w(\eta^T w)\eta^T = 0, \qquad Nu = w(\eta^T u) = w.$$

Thus $N(x)$ is nilpotent for every x, and all three eigenvalues of $V(x) = -aI_3 + N(x)$ equal $-a$. Since $N^2 = 0$,

$$\mathrm{e}^{tV(x)} = \mathrm{e}^{-at}(I_3 + tN(x)).$$

The norm of $N(x)$ is uniformly bounded on $[0, \pi]$; absorbing the linear factor $1 + t\sup_x \|N(x)\|$ into $C_\beta \mathrm{e}^{(a-\beta)t}$ proves the asserted uniform frozen stability.

Finally,

$$u'' + Vu = u'' - au + Nu = u'' - au + (a + \lambda)u - u'' = \lambda u.$$

Also $u'(0) = u'(\pi) = 0$. Hence $U(t, x) = \mathrm{e}^{\lambda t}u(x)$ is a classical Neumann solution. Its norm is $\mathrm{e}^{\lambda t}\|u\|$ in each of the stated spaces, proving unboundedness. □

4 Relation to the source dissipativity hypothesis

For concreteness take $a = \lambda = 1$. At $x = \pi/2$, direct substitution gives

$$V(\pi/2) = \begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ -9 & 0 & -4 \end{pmatrix}.$$

This matrix is not dissipative for any standard ℓ^p-norm. For $1 < p < \infty$, testing the source paper's duality criterion at the first basis vector gives $e_1^T V e_1 = 2 > 0$. At $p = 1$ the first-column criterion would require $2 \leq -9$, and at $p = \infty$ the first-row criterion would require $2 \leq -1$; both fail. Thus the example lies genuinely beyond the paper's dissipative regime, while retaining much stronger pointwise spectral stability than mere nonpositivity of the frozen spectral bounds.

5 Verification, novelty search, and limitations

The accompanying script `code/verify_construction.py` symbolically checks the Gram denominator, $\eta^T u = 1$, $\eta^T w = 0$, $N^2 = 0$, the PDE eigenfunction identity, and the displayed matrix at $x = \pi/2$. These checks are supplementary; all identities are proved above.

A bounded search was performed in the run registry and attempt/solution indexes for arXiv:2208.10324 and the terms "matrix potential", "parabolic system", "pointwise stable", "Hurwitz", and "diffusion instability". An arXiv search used the same close variants and searches for later work by the source authors. It found general Turing and reaction–diffusion instability literature, but no primary source stating this fixed-spectrum, uniformly frozen-stable construction. Novelty confidence is therefore moderate rather than definitive; the algebraic result itself has high confidence.

The theorem does not give a positive boundedness criterion and does not address the source's separate non-elliptic direction. It proves that criteria based only on pointwise spectra—even supplemented by uniform exponential stability of all frozen ODEs—cannot solve the boundedness problem. Human review should focus on the construction of η, the strict positivity in (1), and this scope boundary.

References

[1] A. Dobrick and J. Glück, *Convergence to equilibrium for linear parabolic systems coupled by matrix-valued potentials*, arXiv:2208.10324, version of 6 July 2023.